Essays

|!|BCEEB0200300001|!| ★

Give a brief account on how energy from the Sun is transferred and eventually used by a person to move his/her arms. (11 marks)

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Energy transfer from the Sun to a person (Max. 4)

- Sun light is captured by chlorophyll in plant cells to carry out photosynthesis. (1)
- The light energy is converted to chemical energy / energy stored in organic molecules (e.g. glucose). (1)
- Plant material is consumed by consumers. (1)
- Consumers digest and metabolize the ingested plant material, thus incorporating food substances from plants into their body tissues. (1)
- The chemical energy stored in plants is transferred along the food chain. (1)
- When a person consumes plants / consumers, he/she obtains the chemical energy stored in the organic molecules in food. (1)

Release of energy from food to power movement (Max. 4)

- Organic molecules in food are digested by digestive enzymes in the human body. (1)
- The digested food substances are absorbed in the small intestine and transported to other parts of the body through the bloodstream. (1)
- Muscle cells take up the digested food substances / glucose from the blood. (1)
- Muscle cells carry out respiration to break down glucose, (1)
- which releases energy for the contraction of arm muscles. (1)

Communication (Max. 3)

(11 marks)

##

Skills assessed	Marks
Applying concepts	8
Communication	3

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|!|BCEEA0200500002|!| ★★

✂

Describe how the nitrogen in the protein of a cow may be transferred and incorporated into the protein of a goat living in the same habitat. (11 marks)

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Transfer of nitrogen from a cow to plants (Max. 4)

- The dead body of a cow / Organic waste expelled by a living cow is broken down by fungi and putrefying bacteria. (1)
- The nitrogen in these organic materials is converted to ammonium compounds. (1)
- Under aerobic conditions, the ammonium compounds in the soil are oxidized by nitrifying bacteria to nitrites and then to nitrates. (1)
- Nitrates are absorbed by plants / grasses through their roots. (1)
- Plants / Grasses use nitrates and the intermediates of photosynthesis to form proteins. (1)

Transfer of nitrogen from plants to a goat (Max. 4)

- A goat ingests grasses that contain proteins. (1)
- Proteins from the grasses are digested by proteases in the goat's digestive system into amino acids. (1)
- The amino acids are absorbed in the intestines and transported by blood. (1)
- Body cells take up amino acids by diffusion and active transport. (1)
- Body cells synthesize proteins using the amino acids. (1)

Communication (Max. 3)

(11 marks)

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Skills assessed	Marks
Applying concepts	8
Communication	3

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|!|BCEEA0200800003|!| ★★

Describe how you would measure ***three*** different abiotic factors of a mangrove forest. Further describe how each of these abiotic factors in the ecosystem could adversely affect mangrove plants and explain how the plants have adapted to these conditions.

(12 marks)

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Measuring abiotic factors

- The oxygen concentration in coastal waters can be measured with a dissolved oxygen meter. (1)
- The salinity of coastal waters can be measured with a refractometer. (1)
- The atmospheric temperature can be measured with a thermometer / a temperature sensor connected to a data logger. (1)

Local effects of abiotic factors on mangrove plants

- The soil in mangrove forests are constantly waterlogged and thus poorly aerated, which limits the mangrove plants' oxygen uptake for respiration. (1)
- Mangrove plants are often immersed in saltwater, which could upset the plants' osmotic balance. (1)
- The relatively high temperatures in the ecosystem increase the rate of transpiration in the mangrove plants and may cause desiccation. (1)

Adaptations to the local effects of abiotic factors (Max. 3)

- Mangrove plants grow aerial roots / knee roots that rise above sea level to carry out gas exchange regardless of tide levels. (1)
- Mangrove plants grow lenticels to maximize gas exchange. (1)
- Salt glands on the leaves of certain mangrove plants excrete excess salt, thereby maintaining osmotic balance in the plant. (1)
- Certain mangrove plants have roots that are highly impermeable to salt, thereby maintaining osmotic balance in the plant. (1)
- Mangrove plants generally have thick, waxy cuticles in their leaves to reduce water loss due to transpiration. (1)

Communication (Max. 3)

(12 marks)

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Skills assessed	Marks
Applying concepts	9
Communication	3

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